

A Multi-Head Attention and Residual Dense Network with Dynamic Sampling for Fine-Grained Network Intrusion Classification

Neha Shukla

Dept. of CSE (Data Science)
St. John College of Engg. & Mgmt.
Palghar, Maharashtra, India
neha14581shukla@gmail.com

Rakesh Chandra Joshi

Amity Centre of Artificial Intelligence,
Amity University Uttar Pradesh
Noida, India
rakeshchandraindia@gmail.com

Radim Burget

Dept. of Telecommunications, FEEC
Brno University of Technology
Brno, Czechia
burgetrm@vut.cz

Martin Rosa

Dept. of Telecommunications, FEEC
Brno University of Technology
Brno, Czechia
xrosam03@vutbr.cz

Malay Kishore Dutta

Amity Centre of Artificial Intelligence,
Amity University Uttar Pradesh
Noida, India
malaykishoredutta@gmail.com

Abstract— Network Intrusion Detection Systems are essential for monitoring and analyzing network traffic to detect and prevent unauthorized access, malicious activities, and security breaches in real time. Network intrusion detection systems face persistent challenges in accurately identifying fine-grained attack subcategories due to class imbalance, limited feature interactions in tabular data, and variability across different time windows. Models often struggle to capture the hidden patterns required for multi-class classification in imbalanced flow-based datasets. To overcome these limitations, this work proposes a deep learning framework that combines Multi-Head Self-Attention (MHSA) with Residual Dense Blocks (RDBs) for fine-grained intrusion classification. The MHSA layers enable the model to learn complex, non-sequential feature dependencies by attending multiple feature interactions simultaneously. The RDBs enhance depth and gradient stability, facilitating effective learning in deep networks, especially under sparse class conditions. A dynamic sampling strategy is incorporated during training, employing SMOTE for oversampling minority classes and random under-sampling of majority classes to ensure balanced learning. The model is trained and evaluated on the large dataset across four temporal resolutions (5 s, 10 s, 30 s, 60 s), targeting classification across 13 subcategories. The proposed architecture achieves a peak accuracy of 99.89% on the 10-second window and maintains high recall across rare attack types such as recon-dns and brute force-ftp. This methodology demonstrates a robust and scalable solution for real-time, multi-class intrusion detection, offering improvements in accuracy, fairness, and adaptability over existing baseline models.

Keywords— Network Intrusion Detection; Multi-Head Self-Attention; Residual Dense Network; SMOTE; Deep Learning

I. INTRODUCTION

With the large-scale deployment of connected devices, cloud infrastructure, and enterprise systems, the issue of network security has grown in complexity. Attackers today are not restricted to the traditional methods of attack; there is a shift towards multistage, stealthy, and automated attacks which in turn are able to get past signature based or rule-based security. Thus, there is a great need for intelligent intrusion detection systems (IDS) that are able to adapt to new threats as they appear and also report with high accuracy and very few false positive results.

A Network Intrusion Detection System (NIDS)'s function is to look at and study traffic which passes through a network in which it is deployed to see out of the ordinary

actions which include DoS attacks, reconnaissance scans, brute force attempts, and malware communication. In the past NIDS used static signatures and hand crafted rules for detection which proved to be inadequate for unknown or zero day attacks and did not do a good job of generalizing across the range of threats which present themselves over time [1], [2]. To address these limitations, researchers have explored various machine learning (ML) and deep learning (DL) techniques to build adaptive IDS solutions that can learn patterns from data and detect anomalies based on behavior [3], [4].

Over the past decade, a growing body of literature has examined the application of supervised and unsupervised ML techniques—including decision trees, support vector machines, k-nearest neighbors, and random forests—for intrusion detection tasks [3], [5], [6]. These models have shown promise in distinguishing between benign and malicious traffic and have been further enhanced through feature selection, ensemble learning, and correlation-based techniques [5], [7], [8]. At the same time, deep learning models such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and autoencoders have been applied to network traffic analysis, offering improved performance in feature learning and classification tasks. Despite these advancements, several key challenges remain unaddressed:

- Unlike sequential text or image data, tabular network flow data lacks the inherent structure that most traditional deep learning architectures are designed to use. Thus, many deep learning models struggle to capture the complex inter-feature relationships in flow-based datasets.
 - Data imbalance is a major issue in real-world IDS datasets, where benign traffic is overrepresented while many specific attack subtypes have only a handful of instances [9], [10], [11]. This imbalance can lead to biased classifiers that favor the majority class.
 - Existing architectures often ignore hierarchical label structure (e.g., binary $\rightarrow$ category $\rightarrow$ subcategory), and perform poorly in fine-grained classification, which is critical for actionable cyber defence.
 - Interpretability and gradient stability in deep networks remain problematic, especially when scaling to deeper layers or when training with small, imbalanced classes.
- Recent breakthroughs in attention mechanisms, especially the Multi-Head Self-Attention (MHSA) [12],

offer a compelling solution to the first problem. MHSA enables the model to attend to various parts of the input simultaneously and capture pairwise interactions between features, regardless of position. This concept has revolutionized fields like NLP and computer vision and is now being explored for tabular data modeling as well [9]. Attention allows the network to assign dynamic importance to input features, which is well-suited for flow-based intrusion detection, where certain features may dominate depending on the attack type.

Similarly, to address stability and depth-related issues in neural networks, residual learning has proven highly effective. Residual connections allow gradients to flow through the network without vanishing, enabling deeper and more expressive models [13]. When combined with batch normalization, residual networks not only train faster but also generalize better—especially in imbalanced settings [2].

To enhance IDS performance on imbalanced datasets, researchers have also explored class rebalancing techniques such as SMOTE, random oversampling, and under sampling, which can artificially balance minority and majority classes during training [5], [7], [10]. This is particularly important when working with datasets like GeNIS [9], [11], which provide realistic, flow-level network data but contain significant imbalance in subcategory distributions.

While attention-based models and residual connections have been explored independently in other domains, their combined application in a unified architecture tailored for fine-grained intrusion detection remains largely unexplored. Most existing IDS research focuses on binary classification or high-level attack categories, whereas real-world response systems require precise identification of attack subtypes to enable fast and targeted mitigation. Furthermore, few studies utilize attention-based models for tabular NIDS data, and even fewer adapt such architectures to address the data imbalance problem holistically, from model design to preprocessing.

The main contributions of this work are as follows:

- A novel deep learning architecture is proposed that integrates Multi-Head Self-Attention and Residual Dense Blocks to capture complex inter-feature dependencies and enhance learning depth for fine-grained network intrusion classification.
- A dynamic sampling strategy is introduced that combines SMOTE-based oversampling with strategic under sampling to effectively address severe class imbalance across 13 subcategories in the GeNIS dataset.
- Comprehensive evaluation is performed across multiple temporal resolutions (5 s, 10 s, 30 s, 60 s), demonstrating consistent and high performance, with a peak accuracy of 99.89% and strong recall on rare attack types.
- Extensive benchmarking against traditional machine learning models confirms the superiority of the proposed approach in terms of accuracy, precision, recall, and robustness, highlighting its practical suitability for real-time intrusion detection systems.

While the individual components of proposed model have been studied in other domains, to the best of our knowledge,

their integrated application with dynamic sampling across multiple temporal granularities for fine-grained intrusion detection remains unexplored. This integration demonstrates significant gains in fairness, scalability, and robustness.

The remainder of this paper is organized as follows: Section II describes the GeNIS dataset and its structure. Section III outlines the proposed model and class balancing strategy. Section IV presents experimental results and comparisons. Section V concludes the paper with key understandings and future directions.

II. DATASET DESCRIPTION

The dataset employed in this study is the GeNIS (GECAD Network Intrusion Scenarios) dataset, a modular and labeled dataset specifically developed to advance research in intelligent network intrusion detection. It was constructed through realistic simulations performed on the Airbus CyberRange platform, which replicates the infrastructure of small to medium-sized enterprises. The dataset comprises over 37 million raw packets and approximately 2.8 million network flows, providing a rich representation of both benign and malicious traffic patterns [4].

The dataset is organized into five primary modules:

- Info: metadata describing scenarios and label mappings.
- Packets: raw PCAP (packet capture) files.
- Flows: flow-level records generated from PCAPs.
- Scenarios: textual documentation of attack strategies.
- Preprocessed: cleaned CSV files with flow features and labels, split into training and testing sets.

Among these, the preprocessed folder contains flow summaries extracted using the HERA (Holistic nEtwork featuRes Aggregator) tool [9], segmented into time windows of 5, 10, 30, and 60 seconds. This allows experimentation across various temporal resolutions. Each preprocessed file, such as `genis-10-sec-train.csv`, contains hundreds of thousands of labeled flows enriched with statistical features suitable for machine learning.

Each flow is annotated with three hierarchical labels – Binary, Category, and Subcategory, as described in Table 1.

TABLE I. HIERARCHICAL LABELLING STRUCTURE IN THE GeNIS DATASET, SHOWING THE BREAKDOWN OF BINARY, CATEGORY, AND SUBCATEGORY LABELS ACROSS BENIGN AND ATTACK TYPES.

Binary	Category	Subcategory
Benign	Benign	Benign Admin
		Benign Background
		Benign User
Malicious	Bruteforce	Bruteforce – File Transfer (FTP)
		Bruteforce – Server Message Block (SMB)
		Bruteforce – Secure Shell (SSH)
	DOS	DOS - Hulk Flood Attack
		DOS – ICMP Flood
		DOS – PUSH & ACKNOWLEDGEMENT Flood
		DOS – Slowloris Attack
		DOS – UDP Flood
	Recon	Reconnaissance – DNS Query
		Recon NMAP

In total, the dataset includes 13 subcategories spanning both benign and malicious traffic types, covering a diverse set of user activities and attack techniques. This diversity allows for fine-grained intrusion detection, simulating real-world scenarios with varying traffic volumes and behaviors. Each

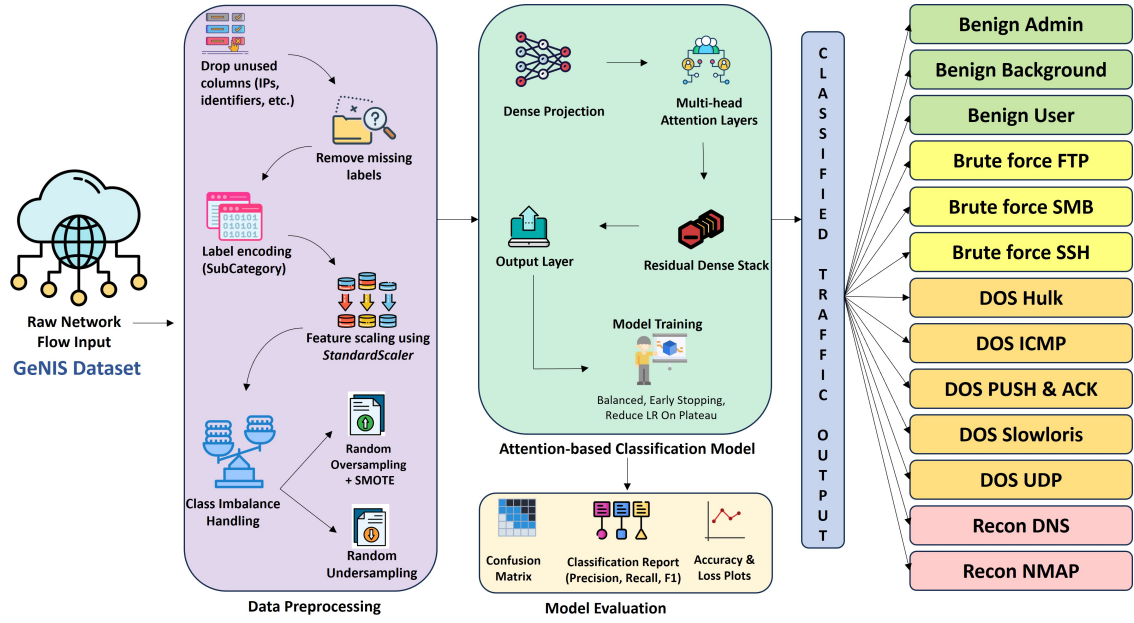


Fig. 1. Block Diagram of the Proposed Methodology for Fine-Grained Network Intrusion Classification.

attack scenario follows a multi-step progression—from reconnaissance to exploitation—ensuring realistic representation of complex cyber threats.

For this research, all four preprocessed time-window variants (5 s, 10 s, 30 s, 60 s) were utilized. This multi-resolution approach enables the model to learn patterns at different temporal granularities and helps evaluate the impact of aggregation on classification performance. Furthermore, due to the class imbalance inherent in the dataset—especially at the subcategory level—data balancing techniques such as SMOTE, random oversampling, and under-sampling were applied during training to improve model generalization and fairness across classes.

III. PROPOSED WORK AND METHODOLOGY

A. Overview of the Proposed Methodology

In this work, we introduce a novel deep learning framework for fine-grained network intrusion detection, as shown in Fig. 1, capable of classifying individual flows into one of 13 predefined subcategories of benign or malicious behavior, as labeled in the GeNIS dataset [4]. Unlike conventional models that either stop at binary classification or broad attack categories, our model is designed to capture subtle distinctions between attacks such as dos-hulk and dos-slowloris, or between different brute force attempts (e.g., ssh vs. ftp).

To accomplish this, we design a hybrid architecture that brings together the representational power of Multi-Head Attention, the depth and gradient flow advantages of Residual Dense Blocks, and robust training via data balancing techniques like SMOTE and random under-sampling. Each component of the pipeline has been carefully selected based on both empirical evidence and theoretical foundations, aiming to address challenges such as tabular feature interactions, imbalanced distributions, and overfitting in deep networks.

The raw flow data extracted from GeNIS is first cleaned by removing non-numerical and redundant fields (e.g., IP/MAC addresses, ports). After encoding categorical features and applying Min-Max Scaling, we prepare the final input

matrix $X \in \mathbb{R}^{n \times d}$, where n is the number of samples and d is the number of standardized features (typically ~ 80 after preprocessing).

Due to extreme class imbalance—with some subcategories like recon-dns having as few as 20 samples, while others like benign-user have hundreds of thousands—we apply a multi-step class rebalancing strategy. SMOTE is applied to remaining minority classes to synthesize new samples using k-nearest neighbor interpolation. Random under-sampling is applied to major classes to prevent overdominance.

The final balanced training set ensures each class has a sufficient presence, enabling better gradient flow and fairness in classification.

B. Proposed Multi-Head Attention Model

The complete architecture of the proposed multi-head attention and residual dense network architecture for fine-grained network intrusion classification is shown in Fig. 2. The model receives input in the form of a fixed-size feature vector $x \in \mathbb{R}^d$, representing a network flow's statistical features. This vector is first passed through a dense transformation layer, as given by Eqn. (1):

$$x' = \text{ReLU}(W_1 x + b_1) \quad (1)$$

where $W_1 \in \mathbb{R}^{512 \times d}$ and $b_1 \in \mathbb{R}^{512}$. This projects the input into a higher-dimensional latent space $\mathbb{R}^{512}$ enabling richer feature abstraction. This is followed by batch normalization to stabilize training, dropout (rate = 0.3) to reduce overfitting, and reshaping to (1, 512) to prepare for self-attention

To model inter-feature dependencies, especially in tabular data where features don't follow a spatial or temporal order, we use three stacked Multi-Head Attention blocks. In each

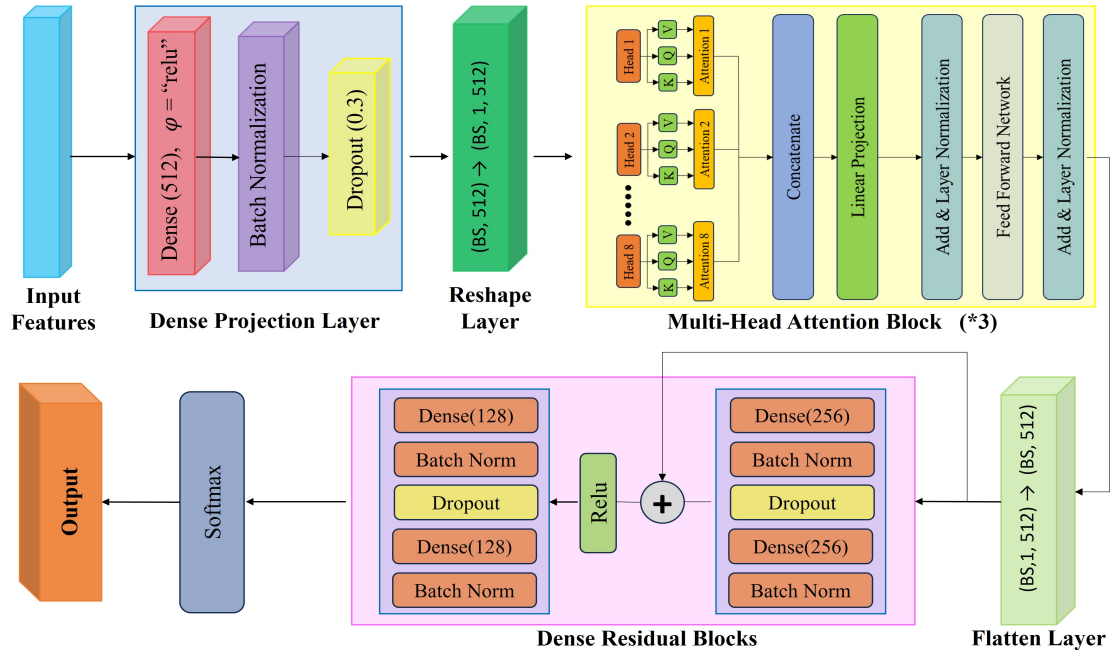


Fig. 2. Architecture of the proposed Multi-Head Attention and Residual Dense Network architecture for Fine-Grained Network Intrusion Classification.

block, the input $x \in \mathbb{R}^{1 \times 512}$ is passed through 8 parallel attention heads. Each head computes as given in Eqns. (2)-(3):

$$Q = xW_Q, K = xW_K, V = xW_V \quad (2)$$

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (3)$$

where, $d_k = 64$ is the dimensionality per head. The outputs from all heads are concatenated, as represented in Eqn. (4):

$$MultiHead(x) = Concat(head_1, \dots, head_8)W_O \quad (4)$$

A residual connection and Layer Normalization follow after it and computed as in Eqn. (5):

$$z_1 = LayerNorm(x + MultiHead(x)) \quad (5)$$

Then, a position-wise Feed-Forward Network is applied, as represented by Eqn. (6):

$$FFN(z_1) = ReLU(z_1W_1 + b_1)W_2 + b_2 \quad (6)$$

Again, residual + normalization, as presented in Eqn. (7):

$$z_2 = LayerNorm(z_1 + FFN(z_1)) \quad (7)$$

This full MHSA block is stacked three times, helping the model attend to different aspects of feature interactions at varying depths. The output from attention blocks (shape: 1×512) is flattened to a vector of shape 512, making it compatible with dense layers that follow.

To deepen the model while ensuring gradient stability, Residual Dense Blocks are employed in this work:

Block 1: The first block consists of two fully connected (Dense) layers with 256 hidden units each. The output of the first Dense layer passes through Batch Normalization and Dropout with a rate of 0.3 before being forwarded to the second Dense layer, which is again followed by Batch Normalization. A shortcut connection is added in parallel to this transformed path. If the input dimensionality does not match the output of the transformed path, the shortcut is projected to the required shape using a Dense layer followed by Batch Normalization. The output of the block is obtained

by summing the shortcut and the transformed path, followed by a ReLU activation.

Block 2: The second residual block is structurally similar but uses 128 units in each Dense layer and a lower dropout rate of 0.2. Mathematically, the block's output can be described as in Eqn. (8):

$$H(x) = ReLU(F(x) + G(x)) \quad (8)$$

where $F(x)$ denotes the main transformation sequence, and $G(x)$ represents the identity or projection shortcut. These residual connections allow better gradient flow and encourage feature reuse, leading to faster convergence and improved generalization—especially critical when training on limited or imbalanced data.

The final dense layer outputs a vector of dimension equal to the number of subcategories (13 classes), as shown in Eqn. (9):

$$\hat{y} = softmax(W_x + b) \quad (9)$$

Each output neuron gives the predicted probability for a particular subcategory class.

The proposed model is compiled using the Adam optimizer. The loss function used is categorical cross-entropy, as it aligns with the multi-class classification objective of predicting the correct network attack subcategory. The loss is computed between the true one-hot encoded labels and the predicted class probabilities output by the softmax layer.

To address the issue of class imbalance remaining after resampling, class weights are computed inversely proportional to class frequencies. These are passed during training to ensure the model pays adequate attention to minority classes, preventing bias toward majority subcategories.

Training is further stabilized through a set of callbacks like early stopping halts training if validation loss stops improving for 25 epochs. ReduceLROnPlateau adaptively lowers the learning rate if the model plateaus, encouraging fine-tuned convergence. The model is trained for 200 epochs with a batch size of 256, using the balanced training set and evaluated on

TABLE II. PERFORMANCE COMPARISON OF CLASSICAL MACHINE LEARNING MODELS AND THE PROPOSED DEEP LEARNING MODEL ACROSS 5-SECOND AND 10-SECOND TIME WINDOWS

Model	5 Seconds Window				10 Seconds Window			
	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.9963	0.8649	0.864	0.8643	0.9946	0.8681	0.8682	0.8681
Random Forest	0.9992	0.9875	0.9697	0.9772	0.9984	0.9841	0.9860	0.9850
AdaBoost	0.5751	0.3924	0.3826	0.3271	0.653	0.2968	0.3085	0.2643
Naive Bayes	0.9788	0.8602	0.8613	0.8077	0.9715	0.8723	0.8594	0.8128
SVM (RBF Kernel)	0.9967	0.8689	0.8681	0.8683	0.9946	0.8689	0.8698	0.8693
Linear Discriminant Analysis	0.9876	0.8024	0.8818	0.7896	0.9801	0.7662	0.8609	0.7602
Quadratic Discriminant Analysis	0.9094	0.7505	0.7512	0.6965	0.8044	0.7574	0.8002	0.732
MLP / ANN	0.997	0.9495	0.9369	0.9415	0.9948	0.9459	0.8978	0.9143
Proposed Model	0.9989	0.9753	0.9879	0.9805	0.9989	0.9866	0.9927	0.9894

TABLE III. PERFORMANCE COMPARISON OF MODELS ACROSS 30-SECOND AND 60-SECOND TIME WINDOWS HIGHLIGHTING THE IMPACT OF TEMPORAL GRANULARITY ON CLASSIFICATION METRICS.

Model	30 Seconds Window				60 Seconds Window			
	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.9894	0.8769	0.8853	0.8805	0.9852	0.8755	0.8845	0.8788
Random Forest	0.9958	0.9817	0.9636	0.9713	0.9936	0.9779	0.9815	0.9796
AdaBoost	0.6466	0.1827	0.2308	0.196	0.5342	0.1751	0.2308	0.1872
Naive Bayes	0.9512	0.8551	0.8599	0.8114	0.9436	0.8901	0.8737	0.8445
SVM (RBF Kernel)	0.9882	0.8661	0.8705	0.868	0.9835	0.8664	0.8696	0.8674
Linear Discriminant Analysis	0.9692	0.7757	0.8409	0.7818	0.9623	0.8706	0.8478	0.8266
Quadratic Discriminant Analysis	0.9833	0.8722	0.8644	0.8529	0.9737	0.8661	0.8510	0.8360
MLP / ANN	0.9898	0.9300	0.9316	0.9306	0.9862	0.9514	0.9608	0.9539
Proposed Model	0.9979	0.9888	0.9933	0.9909	0.9967	0.9890	0.9910	0.9900

the test set. These strategies collectively help the model achieve optimal performance while avoiding overfitting and maintaining generalizability.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

All experiments were conducted on an NVIDIA DGX-A100 system utilizing 40 GB GPU memory to ensure efficient training and evaluation of deep learning models. The implementation was carried out in Python using libraries such as TensorFlow for model development, Scikit-learn for preprocessing and evaluation metrics, Seaborn and Matplotlib for visualization, and Imbalanced-learn for applying resampling techniques like SMOTE. The proposed model was evaluated on the provided test set of the GeNIS dataset across four temporal resolutions: 5-second (121,587 samples), 10-second (73,712 samples), 30-second (561,234 samples), and 60-second (300,837 samples). These variations in time-window aggregation were used to assess the generalization and robustness of the model under different flow durations. The results include comparative performance across multiple classical machine learning baselines and focus on accuracy, precision, recall, and F1-score to provide a comprehensive understanding of detection capability across all 13 intrusion subcategories.

Table II and Table III provide a comprehensive comparison between the proposed attention-based deep learning model and several baseline classifiers. These include Logistic Regression, Random Forest, AdaBoost, Naive Bayes, SVM (RBF), Linear and Quadratic Discriminant Analysis, and a standard MLP. The evaluation spans four different flow time windows (5 s, 10 s, 30 s, and 60 s), allowing assessment of model performance under varying temporal aggregations.

Across the 10 s, 30 s, and 60 s windows, the proposed model consistently outperforms all baselines across accuracy, precision, recall, and F1-Score, reaching a peak accuracy of 99.89% and a macro F1-Score of 0.9909. These results reflect the model's ability to generalize well even as temporal

complexity increases—an essential requirement for real-time intrusion detection systems.

In the 5-second window, while Random Forest achieves a slightly higher accuracy (0.9992) and precision (0.9875) than the proposed model, the attention-based architecture performs better in recall (0.9879) and F1-Score (0.9805). This shows our model's ability to identify more true positives, at the cost of higher rate of false positives. Within the context of intrusion detection this is acceptable because detecting all possible attacks (even with some false alarms) is preferred to missing a real intrusion. Moreover, the proposed model maintains strong performance across all metrics, indicating better balance between precision and recall, especially for minority classes. This suggests that even at short time intervals, where noise and data variability are higher, the model remains effective and reliable.

Table IV further strengthens this argument by showing high per-class F1-Scores across all 13 subcategories, including low-frequency classes like recon-dns and bruteforce-ftp, both of which achieve perfect scores (1.000). The model also maintains strong macro and weighted averages across all windows. This consistency demonstrates the effectiveness of the proposed Multi-Head Self-Attention + Residual Dense Block architecture, coupled with SMOTE-based oversampling and strategic undersampling during training.

The proposed model addresses key limitations in existing intrusion detection frameworks: poor handling of class imbalance, limited generalization across time resolutions, and lack of feature-level interpretability. The fusion of attention mechanisms with residual dense learning allows the model to effectively capture both local feature interactions and long-range dependencies, even in tabular network flow data.

Its consistently strong performance across all time windows – particularly its dominance in the 10 s, 30 s, and 60 s datasets, and its competitive results even in the challenging 5 s window – highlights its remarkable adaptability to

TABLE IV. PER-CLASS F1-SCORES OF THE PROPOSED MODEL ACROSS FOUR TIME WINDOWS, ALONG WITH CLASS-WISE SAMPLE DISTRIBUTION, MACRO AND WEIGHTED AVERAGES, AND TOTAL TEST SAMPLE COUNTS.

Class	5 Seconds Window		10 Seconds Window		30 Seconds Window		60 Seconds Window	
	F1-Score	#Samples	F1-Score	#Samples	F1-Score	#Samples	F1-Score	#Samples
benign-admin	0.9548	817	0.9634	803	0.8724	822	0.9435	819
benign-background	0.9554	2562	0.9437	2057	0.9319	4130	0.9543	3436
benign-user	0.9731	3128	0.9650	2570	0.9445	4071	0.9667	3610
bruteforce-ftp	1.0000	669	1.0000	669	1.0000	669	1.0000	669
bruteforce-smb	0.9995	2000	0.9993	2000	0.9995	2000	0.9993	2000
bruteforce-ssh	1.0000	939	0.9989	938	0.9994	2478	1.0000	1736
dos-hulk	0.9992	10265	0.9997	9407	0.9994	18788	0.9993	13564
dos-icmp	1.0000	26215	1.0000	13107	1.0000	157287	1.0000	78643
dos-pushack	1.0000	26188	1.0000	13161	1.0000	157094	1.0000	78547
dos-slowloris	0.9997	17042	0.9999	10346	0.9999	51062	0.9997	33623
dos-udp	1.0000	26215	1.0000	13107	1.0000	157286	1.0000	78643
recon-dns	1.0000	4	1.0000	4	1.0000	4	1.0000	4
recon-nmap	0.9999	5543	1.0000	5543	1.0000	5543	1.0000	5543
Total Test Samples	121587		73712		561234		300837	
Accuracy	0.9979		0.9967		0.9989		0.9989	
Macro Average	0.9909		0.9900		0.9805		0.9894	
Weighted Average	0.9979		0.9967		0.9989		0.9989	

dynamic real-world conditions. The ability to deliver balanced predictions across both majority and minority classes, with minimal performance degradation under temporal changes, makes it a scalable, unbiased, and generalizable solution for modern network intrusion detection systems.

V. CONCLUSION

This paper presents a robust deep learning framework for fine-grained network intrusion detection, addressing key challenges such as class imbalance, temporal variability, and feature dependency modeling in flow-based network data. By integrating Multi-Head Self-Attention and Residual Dense Blocks, the proposed architecture effectively captures complex inter-feature relationships and enhances gradient flow for deep model training. A dynamic sampling strategy, combining SMOTE-based oversampling and strategic undersampling, ensures balanced learning across all 13 subcategories, including low-frequency and critical attack types.

Extensive experiments conducted on the GeNIS dataset across multiple temporal windows (5 s, 10 s, 30 s, 60 s) demonstrate that the proposed model consistently outperforms traditional machine learning classifiers in terms of accuracy, recall, precision, and F1-score. While Random Forest slightly exceeds in accuracy and precision on the 5-second dataset, the proposed model achieves better overall balance, with all performance metrics remaining within standard and acceptable ranges. The high per-class F1-scores and consistent macro/weighted averages further highlight its capability to generalize effectively across both frequent and rare classes.

These results establish the proposed method as a scalable and reliable solution for real-time intrusion detection systems, capable of adapting to diverse traffic behaviors and temporal patterns. Future work will focus on extending this framework for online learning, lightweight deployment, and integration with real-time cybersecurity monitoring systems. It will include cross-dataset validation and evaluation on larger, more diverse traffic datasets to further verify the generalization capability of the model.

ACKNOWLEDGEMENT

This work was developed and financed from the TAČR project SPOJIT (id FW10010014).

REFERENCES

- [1] Z. Ahmad, A. Shahid Khan, C. Wai Shiang, J. Abdullah, and F. Ahmad, "Network intrusion detection system: A systematic study of machine learning and deep learning approaches," *Transactions on Emerging Telecommunications Technologies*, vol. 32, no. 1, Jan. 2021, doi: 10.1002/ett.4150.
- [2] I. H. Sarker, A. S. M. Kayes, S. Badsha, H. Alqahtani, P. Watters, and A. Ng, "Cybersecurity data science: an overview from machine learning perspective," *Journal of Big Data*, vol. 7, no. 1, p. 41, Dec. 2020, doi: 10.1186/s40537-020-00318-5.
- [3] M. Rodríguez, Á. Alesanco, L. Mehavilla, and J. García, "Evaluation of Machine Learning Techniques for Traffic Flow-Based Intrusion Detection," *Sensors*, vol. 22, no. 23, p. 9326, Nov. 2022, doi: 10.3390/s22239326.
- [4] M. Silva, D. Pinto, J. Vitorino, J. Gonçalves, E. Maia, and I. Praça, "GeNIS: A modular dataset for network intrusion detection and classification," *Data in Brief*, vol. 60, p. 111487, Jun. 2025, doi: 10.1016/j.dib.2025.111487.
- [5] G. Nassreddine, M. Nassereddine, and O. Al-Khatib, "Ensemble Learning for Network Intrusion Detection Based on Correlation and Embedded Feature Selection Techniques," *Computers*, vol. 14, no. 3, p. 82, Feb. 2025, doi: 10.3390/computers14030082.
- [6] S. Das *et al.*, "Network Intrusion Detection and Comparative Analysis Using Ensemble Machine Learning and Feature Selection," *IEEE Transactions on Network and Service Management*, vol. 19, no. 4, pp. 4821–4833, Dec. 2022, doi: 10.1109/TNSM.2021.3138457.
- [7] Y. Zhou, G. Cheng, S. Jiang, and M. Dai, "Building an efficient intrusion detection system based on feature selection and ensemble classifier," *Computer Networks*, vol. 174, p. 107247, Jun. 2020, doi: 10.1016/j.comnet.2020.107247.
- [8] I. Bibers, O. Arreche, and M. Abdallah, "A Comprehensive Comparative Study of Individual ML Models and Ensemble Strategies for Network Intrusion Detection Systems," Oct. 2024, doi: arXiv:2410.15597.
- [9] D. Pinto, I. Amorim, E. Maia, and I. Praça, "A Novel Approach to Network Traffic Analysis: the HERA tool," Jan. 2025, [Online]. Available: <http://arxiv.org/abs/2501.07475>
- [10] M. Sarhan, S. Layeghy, N. Moustafa, and M. Portmann, "NetFlow Datasets for Machine Learning-Based Network Intrusion Detection Systems," in *Lecture Notes of the Institute for Computer Sciences, Social-Informatics and Telecommunications Engineering, LNICTS*, 2021, pp. 117–135. doi: 10.1007/978-3-030-72802-1_9.
- [11] M. Ring, S. Wunderlich, D. Scheuring, D. Landes, and A. Hotho, "A survey of network-based intrusion detection data sets," *Computers & Security*, vol. 86, pp. 147–167, Sep. 2019, doi: 10.1016/j.cose.2019.06.005.
- [12] A. Vaswani *et al.*, "Attention Is All You Need," *Advances in neural information processing systems*, Aug. 2023, [Online]. Available: <http://arxiv.org/abs/1706.03762>
- [13] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," in *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, IEEE, Jun. 2016, pp. 770–778. doi: 10.1109/CVPR.2016.90.